

Name: Janhvi Sawarkar

PRN: 202501040125

Batch: CS2(C21)

CODETANTRA LAB PRACTICAL 4

CODETANTRA

Home Learn Anywhere

202501040125@mitaoe.ac.in Support Logout

4.1.1. Pandas - series creation and manipulation19:57

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

Submit

1import pandas as pd

2

3# Take inputs from the user to create a list of numbers

4numbers = list(map(int, input().split()))

5

6# Create a Pandas series from the list of numbers

7s = pd.Series(numbers)

8# Grouping by even and odd numbers and calculating the mean

9grouped = s.groupby(s%2==0).mean()

10

11# Display the mean of even and odd numbers with labels

12grouped.index = ['Even' if is_even else 'Odd' for is_even in

13grouped.index]

14print("Mean of even and odd numbers:")

15print(grouped)

Average time0.010 sMaximum time0.022 s

9.83 ms22.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 122 msDebug

Expected output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd 5.0

Even 6.0

dtype: float64

Actual output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd 5.0

Even 6.0

dtype: float64

TerminalTest cases

PrevResetSubmitNext

CODETANTRA

Home Learn Anywhere

202501040125@mitaoe.ac.in Support Logout

4.1.2. Dictionary to dataframe43:43

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.

datafram...

Submit

1import pandas as pd

2

3# Provided dictionary of lists

4data = {

5 'Name': ['Alice', 'Bob', 'Charlie'],

6 'Age': [25, 30, 35],

7}

8

9# Convert the dictionary to a DataFrame

10df = pd.DataFrame(data)

11

12# Display the original DataFrame

13print("Original DataFrame:")

14print(df)

15

Average time0.081 sMaximum time0.083 s

81.00 ms83.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 179 msDebug

Expected output

Original DataFrame:

..... Name Age

0 Alice 25

1 Bob 30

2 Charlie 35

New name: Susan

Actual output

Original DataFrame:

..... Name Age

0 Alice 25

1 Bob 30

2 Charlie 35

New name: Susan

TerminalTest cases

PrevResetSubmitNext

CODETANTRAHomeLearn Anywhere202501040125@mitaoe.ac.inSupportLogout

4.1.3. Student Information15:33

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases+

studentin...

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])
6
7 # write your code here..
8 print("First five rows:")
9 print(data.head())
10 av=round(data["Age"].mean(),2)
11 print("Average age:",av)
12 print("Students with a grade up to B")
13 fd=data[data["Grade"].isin(['A','B'])]
14 print(fd)
```

Average time0.038 s38.00 msMaximum time0.056 s56.00 ms

1 out of 1 shown test case(s) passed1 out of 1 hidden test case(s) passed

Test case 145msDebug

Expected output

studentdata.txt

First five rows:

-- Name - Age - Grade

0 - John - 25 - A

1 - Allin - 22 - B

2 - Emma - 24 - A

Actual output

studentdata.txt

First five rows:

-- Name - Age - Grade

0 - John - 25 - A

1 - Allin - 22 - B

2 - Emma - 24 - A

TerminalTest cases

< PrevResetSubmitNext>

CODETANTRAHomeLearn Anywhere202501040125@mitaoe.ac.inSupportLogout

4.2.1. Month with the Highest Total Sales14:29

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases+

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8 #convert date column to datetime
9 df['Date']=pd.to_datetime(df['Date'])
10 #extract year-month for grouping
11 df['Month']=df['Date'].dt.to_period('M')
12 #calculate total sales for each row(quantity * price)
13 df['Total_Sales']=df['Quantity']*df['Price']
14 #group by month and sum the total sales
15 monthly_sales=df.groupby('Month')['Total_Sales'].sum()
```

Average time0.035 s35.00 msMaximum time0.045 s45.00 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 145msDebug

Expected output

sales_data.csv

Best month: 2025-01

Total sales: \$1210.00

Actual output

sales_data.csv

Best month: 2025-01

Total sales: \$1210.00

TerminalTest cases

< PrevResetSubmitNext>

CODETANTRA

HomeLearn Anywhere

202501040125@mitaoe.ac.inSupportLogout

4.2.2. Best Selling Product

16:15

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

1import pandas as pd

2

3# Prompt the user for the file name

4file_name = input()

5

6# Load the data

7df = pd.read_csv(file_name)

8

9ps=df.groupby('Product')['Quantity'].sum()

10# Find the product with the highest total quantity sold

11best_product =ps.idxmax()

12highest_quantity = ps.max()

13

14# Display the result

15print(f"Best selling product: {best_product}")

Average time0.014 sMaximum time0.015 s

13.67 ms16.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 115 msDebug

Expected outputActual output

sales_data.csvsales_data.csv

Best selling product: Product ABest selling product: Product A

Total quantity sold: 23Total quantity sold: 23

TerminalTest cases

PrevResetSubmitNext

CODETANTRA

HomeLearn Anywhere

202501040125@mitaoe.ac.inSupportLogout

4.2.3. City that Sold the Most Products

03:50

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

1import pandas as pd

2

3# Prompt the user for the file name

4file_name = input()

5

6# Load the data

7df = pd.read_csv(file_name)

8

9# write the code.

10a=df.groupby('City')['Quantity'].sum()

11best_city=a.idxmax()

12

13# Display the result

14print(f"City sold the most products: {best_city}")

Average time0.014 sMaximum time0.014 s

13.67 ms14.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 114 msDebug

Expected outputActual output

sales_data.csvsales_data.csv

City sold the most products: Los AngelesCity sold the most products: Los Angeles

TerminalTest cases

PrevResetSubmitNext

CODETANTRAHomeLearn Anywhere202501040125@mitaoe.ac.inSupportLogout

4.2.4. Most Frequently Sold Product Pairs18:30

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C: Appears in transactions on 2025-01-04.

frequentl...Submit

```
1 import pandas as pd
2 from itertools import combinations
3 from collections import Counter
4
5 # Prompt user to input the file name
6 file_name = input()
7
8 # Read data from the specified CSV file
9 df = pd.read_csv(file_name)
10
11 # write the code
12 count_pairs=Counter()
13
14 for date,group in df.groupby('Date'):
15     products=group['Product']
```

Average time0.016 s15.67 msMaximum time0.018 s18.00 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1418 msDebug

Expected outputActual output

sales_data.csvsales_data.csv

Product A and Product B: 2 timesProduct A and Product B: 2 times

Product A and Product C: 2 timesProduct A and Product C: 2 times

TerminalTest cases

< PrevResetSubmitNext>

CODETANTRAHomeLearn Anywhere202501040125@mitaoe.ac.inSupportLogout

4.2.5. Titanic Dataset Analysis and Data Cleaning14:22

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

- Display the first 5 rows of the dataset.
- Display the last 5 rows of the dataset.
- Get the shape of the dataset (number of rows and columns).
- Get a summary of the dataset (using `.info()`).
- Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
- Check for missing values and display the count of missing values for each column.
- Fill missing values in the 'Age' column with the median age.
- Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
- Drop the 'Cabin' column due to many missing values.
- Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101262,7.925,,S
```

titanicDat...Submit

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # 1. Display the first 5 rows of the dataset
8 print(data.head(5))
9
10 # 2. Display the last 5 rows of the dataset
11 print(data.tail(5))
12
13 # 3. Get the shape of the dataset
14 print(data.shape)
15
```

Average time0.152 s152.00 msMaximum time0.152 s152.00 ms

1 out of 1 shown test case(s) passed

Test case 14152 msDebug

Expected outputActual output

PassengerId Survived Pclass Fare Cabin Embarked PassengerId Survived Pclass Fare Cabin Embarked

0 1 0 7.2500 0 3 7.2500 0 NaN S

1 1 0 71.2833 1 1 71.2833 1 C85 C

2 1 0 7.9250 2 3 7.9250 2 NaN S

TerminalTest cases

< PrevResetSubmitNext>

CODETANTRA

HomeLearn Anywhere

202501040125@mitaoe.ac.inSupportLogout

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- Convert the 'Sex' column to numeric values (male: 0, female: 1).
- One-hot encode the 'Embarked' column, dropping the first category.
- Get the mean age of passengers.
- Get the median fare of passengers.
- Get the number of passengers by class.
- Get the number of passengers by gender.
- Get the number of passengers by survival status.
- Calculate the survival rate of passengers.
- Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, S
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
```

Explorer

titanicDat...

Submit

1import pandas as pd
2import numpy as np
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6data['FamilySize'] = data['SibSp'] + data['Parch']
7
8# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
9
10data['IsAlone'] = np.where(data['FamilySize'] == 0, 1, 0)
11
12# 2. Convert 'Sex' to numeric (male: 0, female: 1)
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14
15# 3. One-hot encode the 'Embarked' column

Average time0.026 sMaximum time0.026 s1 out of 1 shown test case(s) passed

Test case 126 msDebug

Expected outputActual output
29.6991176470588229.69911764705882
14.454214.4542
3---4913---491
1---2161---216
2---1842---184
Name: Pclass, dtype: int64Name: Pclass, dtype: int64

TerminalTest cases

PrevResetSubmitNext

CODETANTRA

HomeLearn Anywhere

202501040125@mitaoe.ac.inSupportLogout

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Calculate the survival rate by class.
- Calculate the survival rate by embarkation location (Embarked_S).
- Calculate the survival rate by family size (FamilySize).
- Calculate the survival rate by being alone (IsAlone).
- Get the average fare by passenger class (Pclass).
- Get the average age by passenger class (Pclass).
- Get the average age by survival status (Survived).
- Get the average fare by survival status (Survived).
- Get the number of survivors by class (Pclass).
- Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, S
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
```

Explorer

titanicDat...

Submit

1import pandas as pd
2import numpy as np
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6data['FamilySize'] = data['SibSp'] + data['Parch']
7data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10# 1. Calculate the survival rate by class
11print(data.groupby('Pclass')['Survived'].mean())
12
13# 2. Calculate the survival rate by embarked location
14print(data.groupby('Embarked_S')['Survived'].mean())
15

Average time0.033 sMaximum time0.033 s1 out of 1 shown test case(s) passed

Test case 163 msDebug

Expected outputActual output
PclassPclass
1---0.6296301---0.629630
2---0.4728262---0.472826
3---0.2423633---0.242363
Name: Survived, dtype: float64Name: Survived, dtype: float64
Embarked_SEmbarked_S

TerminalTest cases

PrevResetSubmitNext

CODETANTRA

HomeLearn Anywhere

202501040125@mitaoe.ac.inSupportLogout

4.2.8. Titanic Dataset Analysis and Data Cleaning - 414/12

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
```

TitanicDat...

Submit

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
7
8
9 # 1. Get the number of survivors by gender
10 print(data[data['Survived']==1]['Sex'].value_counts())
11
12 # 2. Get the number of non-survivors by gender
13 print(data[data['Survived']==0]['Sex'].value_counts())
14
15 # 3. Get the number of survivors by embarked location
```

Average time0.027 s27.00 ms

Maximum time0.027 s27.00 ms

1 out of 1 shown test case(s) passed

Test case 127msDebug

Expected output

Actual output

female.....233

female.....233

male.....109

male.....109

Name: Sex, dtype: int64

Name: Sex, dtype: int64

male.....468

male.....468

female.....81

female.....81

Name: Sex, dtype: int64

Name: Sex, dtype: int64

TerminalTest cases

< Prev

Reset

Submit

Next >